

MECHANICAL ENGINEERING DRAWING*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the loose sheet and questions of Group-B must be written in drawing sheet. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

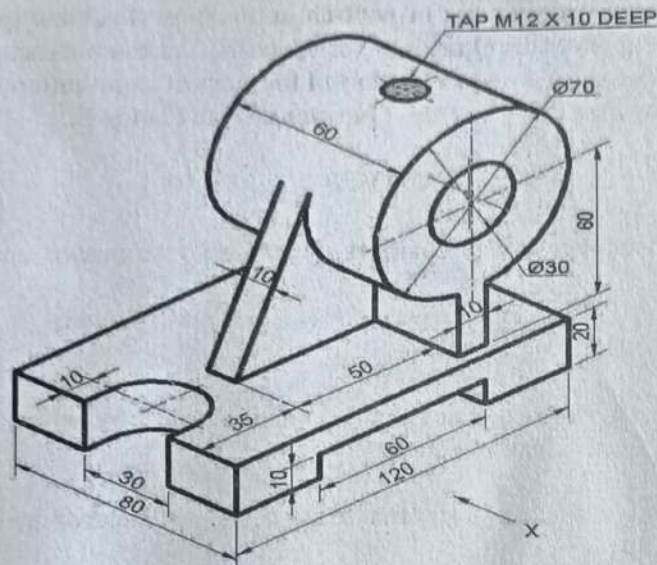
1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10
 - I. Detailed drawing of each part of a machine is called - (a) Assembly drawing, (b) part drawing, (c) patent drawing, (d) tabular drawing
 - II. Tolerance is the difference between- (a) maximum and minimum size (b) maximum and basic size (c) basic size and minimum size (d) none of these
 - III. Sectional views reveal- a) Inner details, b) external features, c) overall size of the object d) none of these.
 - IV. Basic Hole means- (a) Upper deviation is zero, (b) lower deviation is zero, (c) both a & b, (d) none of these
 - V. H7g6- what type of fit is this? a) Clearance b) interference c) transition d) none of these.
 - VI. In a full sectional view the object is imagined to be cut off (a) one-third (b) one half (c) one fourth.
 - VII. A knuckle joint is a mechanical joint used to connect two rods which are under (a) tensile load (b) Compressive load (c) shear load (d) none of these
 - VIII. When a horizontal cylinder penetrates a vertical cone the intersecting surface will have appeared as (a) a straight line (b) curve (c) None of these
 - IX. A coupling which permits within certain limits relative rotation and variation in the alignment of shafts is called (a) Flexible coupling (b) Rigid coupling (c) Universal coupling (d) None of these
 - X. Radial-line development method are used to develop (a) Cylinder (b) Cone (c) Pyramid (d) both b & c
 - XI. Which of the following are part of the knuckle joint (a) pin (b) fork (c) eye (d) all of this
 - XII. Minimum no. of views necessary to determine the shape of the object are: (a) 1, (b) 2, (c) 3, (d) 4
 - XIII. Most common material for pulley is (a) cast iron (b) wood (c) rubber (d) brass
 - XIV. The angle between isometric axes is- (a) 30° , (b) 45° , (c) 90° , (d) 120°
 - XV. In 1st angle projection, top view always— (a) above the front view, (b) below the front view, (c) left side of the front view, (d) right side of the front view.

GROUP-B

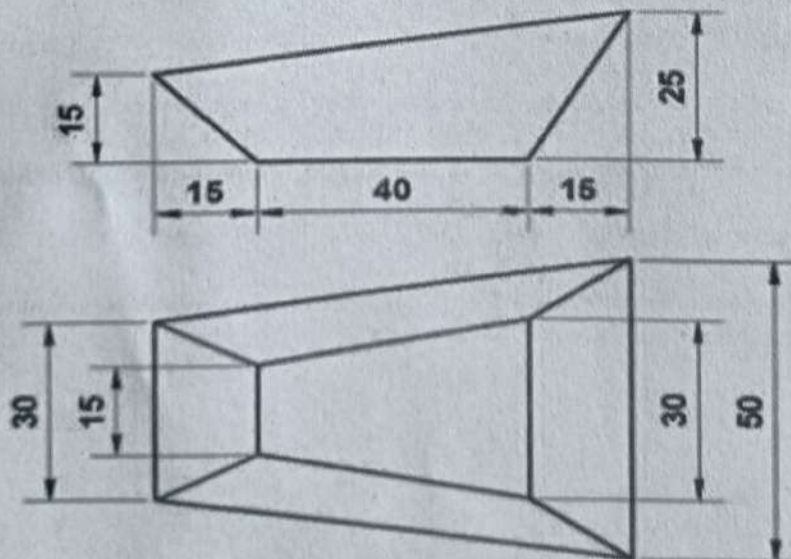
(Answer any Five (05) questions.) (8 x 5 = 40 marks)

2. Draw a full sectional front and top view of following diagram.

(4+4 = 8)

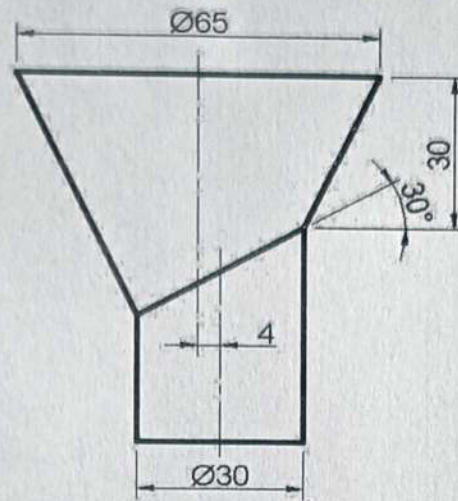


3. An oblique pentagonal prism, having base 30 mm side and height 70 mm long axis, has the axis inclined at 60° to the base and is resting on its base on HP such that one of the rectangular faces is parallel to the VP. Draw the development of its lateral surfaces. (8)
4. A vertical cone, base diameter 75 mm and axis 100 mm long, is completely penetrated by a cylinder of 45 mm diameter. The axis of the cylinder is parallel to HP and VP and intersect axis of cone at a point of 25 mm above from base. Draw projections showing curves of intersection. (8)
5. The Plan and Elevation of a tray are as shown in the figure below. Develop the surface of the tray. (8)



6. Develop the lateral surfaces of these transition pieces.

(4+4= 8)



7. Draw only the sectional elevation of a socket & spigot joint considering the following dimensions:

(8)

- Dimensions of both rods = 25 mm
- Diameter of collar of spigot = 50 mm
- Thickness of collar of spigot = 15 mm
- Outside diameter of spigot = 32 mm
- Length of spigot after the collar = 85 mm
- Diameter of collar of socket = 65 mm
- Thickness of collar of socket = 25 mm
- Inside diameter of socket = 32 mm
- Length of cotter = 100 mm
- Thickness of cotter = 6 mm
- Taper of cotter = 1 in 30

8. Draw full sectional front view of a protected type rigid flange coupling having diameter of the shaft diameter $D=40$ mm. Take other dimensions from the standard proportion are also given below

(8)

- Length of hub = $1.5D$
- Outside hub diameter = $2D$
- Flange diameter = $4D$
- Flange width = $0.8D$
- Flange thickness = $0.5D$
- Thickness of the protective circumferential flange = $0.25D$
- Number of bolts = 4
- Pitch circle diameter of the bolts = $2.8D$

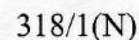
Missing dimension may be assumed (if any).

9. Draw neatly and proportionately a spur gear of 24 teeth, P.C.D. = 300 mm. pressure angle = 20° , showing at least three number of teeth profile.

(8)

10. a) Define tolerance. What is mean by Hole basis and Shaft basis system. b) write short notes (any two)
i) clearance fit ii) interference fit iii) transition fit.

[(1+3)+(2+2)]



3200682

Time Allowed: 3 Hours

Full Marks: 60

Answer any Five (05) Questions from Group-B.

GROUP-A

-



3200682

- (xii) For development of lateral surface of a open-ended right-circular cylinder of radius 50 mm and height 80 mm, the size of sheet-metal requirement (in mm^2) will be (a) 2π (b) π (c) 0.2π (d) 0.02π .
- (xiii) Detailed drawing of each part of a machine is called (a) part drawing (b) assembly drawing (c) patent drawing (d) tabular drawing
- (xiv) When the line of intersection formed by two solids is curve, then the solids are (a) pyramid vs. cuboids (b) prism vs. cube (c) pyramid vs. cube (d) Cone vs. Cylinder
- (xv) Bill of Materials is a part of (a) Part drawing (b) assembly drawing (c) blue-print (d) nowhere.

GROUP-B

All drawings should be done following first-angle projection method only.

2. A square prism of side of base 40 mm and axis 70 mm is resting on its base on H.P., such that one rectangular face of it is parallel to V.P. It is cut by a section-plane, perpendicular to V.P. and inclined at 45° to H.P. The section-plane is passing through the top of an edge perpendicular to V.P. Draw the development of the lateral surfaces of the truncated prism, assuming suitable scale. [8]
3. An air-conditioning duct connects a circular section of 40 mm diameter with a square section of 70 mm side through a 75 mm long transition piece. Draw the development of the lateral surface of the transition piece. Both sections of the transition pieces are coaxial and it is resting on its square base on the HP. [8]



4. A vertical cylinder of diameter 70 mm is completely penetrated by another horizontal cylinder of diameter 50 mm, their axes bisecting each other at right angles. Draw their projections showing curves of intersection, assuming the axis of the penetrating cylinder is parallel to parallel to both H.P. and V.P. Height/ length of the cylinders may be assumed suitably. [8]

5. According to the relative position of the tolerance zones of coupled parts, classify fit. In a limit system, the following limits are specified for the mating parts.

- 0.005

+ 0.020

Shaft: $30^{-0.018}$ mm

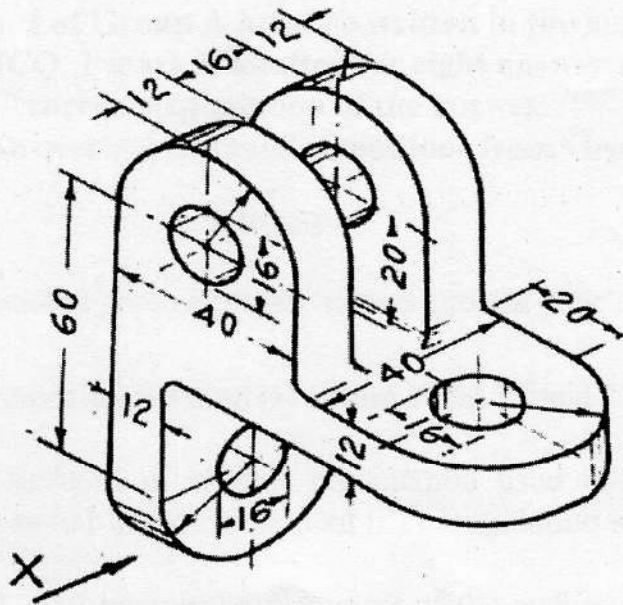
Hole: $30^{-0.000}$ mm

Determine (a) basic size (b) shaft and hole tolerances (c) the shaft and hole limits (d) type of fit.

[2 + (1+2+2+1) = 8]



6. Draw the elevation viewed from X and top view as given in figure below. All dimensions are in mm.



[8]

7. Classify assembly drawings based on its use. Prepare the Bill of materials separately for both 'Knuckle Joint' and 'Screw Jack'.
[3 + (2 + 3) = 8]



8. Draw the half-sectional front view and side view of a 'Protected-type Flange Coupling' having shaft diameter (D) = 50 mm. Some of the standard proportions are also given below:

- (a) Length of the hub = $1.5D$,
- (b) Outside hub diameter = $2D$,
- (c) Flange diameter = $4D$,
- (d) Flange width = $0.8D$,
- (e) Flange thickness = $0.5D$,
- (f) Thickness of the protective circumferential flange = $0.25D$,
- (g) Number of bolts = 4,
- (h) Pitch circle diameter of the bolts = $2.8D$.

Missing dimensions, if any, may be assumed.

[8]

9. Construct the involute tooth profile of a spur gear, showing at least three numbers teeth profile, having 30 numbers of teeth, 300 mm pitch-circle diameter and pressure angle of 20° .
[8]

[NEATNESS IS HIGHLY RECOMMENDED]

MECHANICAL ENGINEERING DRAWING*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 is compulsory, this answer is to be made in separate loose answer script(s) provided for the purpose. The loose sheet must be stapled with drawing sheet at the end of examination. In Question No. 1 (Group – A), out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for explanation of the answer.

&

Answer any Four (04) Questions from the rest (Group – B).

GROUP-A

Answer any Ten (10) questions from Question No. 1

1. Choose the correct answer from the given alternatives and explain your answer (any ten): **2 x 10 = 20**
 - i) Knuckle joint is more rigid than socket and spigot joint. (a) True or (b) False
 - ii) Which development method is used for development of cone ? - (a) Parallel line method, (b) Approximation method, (c) Radial line method, (d) none of these
 - iii) When a plane is parallel to a reference plane, its projection on that plane is – (a) a line, (b) a plane, (c) a solid, (d) none of these.
 - iv) When a clockwise rotation of a nut moves towards the bolt head, the thread is said to be left hand thread. (a) True or (b) False
 - v) When two intersecting cylinders are of same diameters, then the lines of intersection will be – (a) two straight line, (b) one straight line, (c) one circle (d) none of these.
 - vi) In spur gears – (a) both shafts are parallel, (b) teeth are straight, (c) teeth are parallel to axis, (d) all of these.
 - vii) In spur gear, addendum is the radial height of a tooth above the – (a) addendum circle, (b) dedendum circle, (c) pitch circle, (d) none of these.
 - viii) In hole basis system – (a) lower deviation of the hole will be zero, (b) upper deviation of the hole will be zero, (c) both (a) & (b), (d) none of these.
 - ix) Crowning of a pulley is done to – (a) increase strength of the pulley, (b) avoid slipping of the belt, (c) prevent the wear and tear of the belt, (d) all of these.
 - x) The device used to connect two shafts together at their ends for transmission of power is known as – (a) gear, (b) coupling, (c) pulley, (d) all of these.
 - xi) The joint commonly used to convert rotary motion into reciprocating motion or vice versa is – (a) gib & cotter joint, (b) knuckle joint, (c) socket & spigot joint, (d) sleeve & cotter joint.
 - xii) Which one is the female part of socket & spigot joint? – (a) spigot, (b) socket, (c) cotter, (d) none of these.
 - xiii) An intentional difference between the hole dimension and shaft dimension for any type of fit is called – (a) limits, (b) allowance, (c) tolerance, (d) none of these
 - xiv) In the hole and shaft pair size $\Phi 50H7/c8$, the number 7 & 8 indicate – (a) tolerance grade, (b) accuracy, (c) surface texture, (d) none of these.

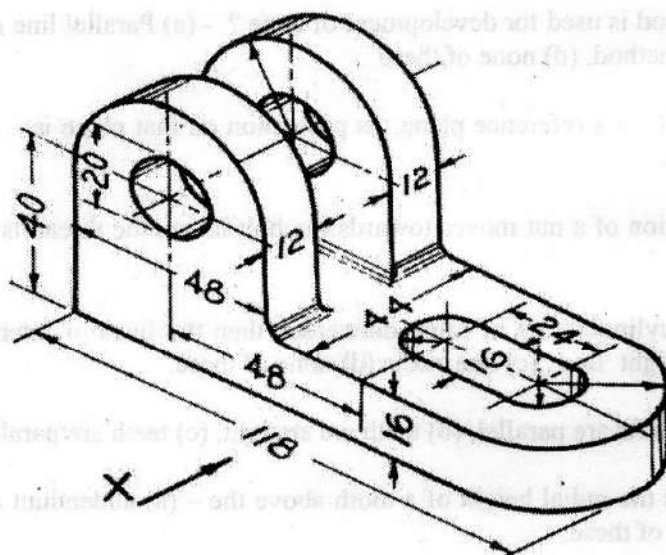
- xv) The development sketch for the lateral surface of a cone involves – (a) an arc of length equal to the radius of the base, (b) an arc of length equal to the circumference of the base, (c) a cord of length equal to the circumference of the base, (d) an arc of length equal to the true length of the slant generator.

GROUP-B

Answer any Four (04) questions.

4 x 10 = 40

2. Draw the development of an oblique cone of diameter 50 mm and axis 70 mm long, having the axis inclined at 60° to the base is resting on HP on its base with its axis parallel to V.P. 10
3. A vertical cylinder of 50 mm diameter and height 75 mm is resting on its base on horizontal plane is completely penetrate by another similar cylinder (same diameter and length). Their axes bisect each other at right angle and parallel to V.P . Draw their projection showing lines of interpenetration on the two cylinders. 10
4. Draw front view from 'X' and top view (with dimension) of the object of which isometric view is given below. All dimensions are in mm. 10

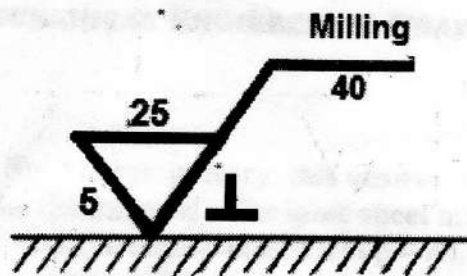


5. Draw neatly and proportionately a spur gear of 30 teeth , P.C.D = 300 mm , Pressure angle = 20° , showing at least three numbers teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle etc. 10
6. Draw only front view (elevation) of a knuckle joint having the pin 25 mm diameter. Assume the dimensions of other parts of the knuckle joint as per proportions. 10
7. Draw a sectional front and top view of a socket & spigot joint where inner diameter of the pipe is 150 mm and Maximum outer diameter is 280mm. Assume suitable dimensions for other parameters. 10
8. Draw half-sectional front view and side view of a protected type flange coupling having shaft diameter (D) = 50 mm. Some of the standard proportions are also given below:
 - a. Outside hub diameter = 2D,
 - b. Length of the hub = 1.5D,
 - c. Flange diameter = 4D,
 - d. Width of the flange = 0.8D,
 - e. Thickness of the flange = 0.5D,
 - f. Number of bolts = 4
 Missing dimensions, if any, may be assumed.

10

9. a) State the meaning of the figure shown in below -

1 x 1



b) Draw the symbols for the following features which are controlled in geometrical tolerance :

1 x 1

- i) Flatness
- ii) Cylindricity.
- iii) Angularity
- iv) Parallelism
- v) Concentricity

2909

June 2022

318/1(N)

MECHANICAL ENGINEERING DRAWING*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. Lateral surface of a cone can be developed by -
i) parallel line development method, ii) conical method iii) radial line development method
iv) approximate method
- ii. Which of the followings is/are never shown in sections? -
(a) Rivets (b) keys (c) bolts (d) all of these
- iii. In conventional representation of Cast Iron, the angle of hatching line with the longitudinal axis is -
(A) 0 degree, (B) 35 degree, (C) 45 degree, (D) 90 degree.
- iv. Detailed drawing of each part of a machine is called -
(a) Assembly drawing, (b) part drawing, (c) patent drawing, (d) tabular drawing
- v. Tolerance is the difference between- (a) maximum and minimum size (b) maximum and basic size
(c) basic size and minimum size (d) none of these
- vi. In a steam engine, the valve rod is connected to an eccentric by means of a -
(a) knuckle joint, (b) universal joint (c) flange coupling (d) cotter joint.
- vii. Following method is used for Development of Right Cylinder - (A) Radial line method (B) Parallel line method (C) Approximate Method (D) All of the above
- viii. When line of intersection formed by two solids is curve then the solids are-
(a) Cone, cylinder (b) Cube, Prism (c) pyramid, cube (d) pyramid, cuboids.
- ix. Sectional views reveal-
a) inner details, b) external features, c) overall size of the object, d) none of these
- x. The value of the ratio of isometric length to true length is -
a) 0.141, b) 0.372, c) 0.815, d) 0.642.
- xi. Blue prints are made for - a) working drawing assembly, b) designed assembly, c) installation assembly d) sub-assembly drawing
- xii. When the cutting plane cuts the entire object the section is known as-
(a) full section (b) half section (c) revolved section (d) removed section.
- xiii. Which type of solid has two bases of parallel and equal polygon?
(A) Pyramid, (B) Cone, (C) Cylinder, (D) Prism
- xiv. Basic Hole means-
(a) Upper deviation is zero, (b) lower deviation is zero, (c) both (a) & (b), (d) none of these
- xv. Standard tapers of a cotter used in industrial practice is -
a) 1 in 50, b) 1 in 30, c) 1 in 100, d) all of the above

2. Fill in the blanks (any ten):

1x10

- a) Base of an oblique cone is a _____
- b) Bill of materials is a part of _____ drawing.
- c) The full form of BIS _____
- d) LMC stands for _____.
- e) Most common material for pulley is _____.
- f) The development of a right prism will be _____.
- g) Hidden lines are used to show _____ edges of the object.
- h) The development of lateral surface of a pyramid consists of a number of equal _____ triangle in contact.
- i) To show the surface of section, hatching lines are drawing at _____°.
- j) Spigot is a part of _____ joint.
- k) Angle between the isometric axes is _____.
- l) Dedendum is the radial depth of a tooth below the _____ circle.
- m) Curves of penetration are found when _____.
- n) Symbol used to represent single sided fillet weld is _____.
- o) Intersection of one plane surface and a curved surface is always a _____ line.

3. Answer the following questions (any ten):

1x10

- a) Show the symbol of CI section in Engineering Drawing.
- b) H7 g6 – what type of fit is this?
- c) State the ratio of length to end width of an arrow.
- d) Define cutting plane of a sectional view.
- e) Distinguish between cotter joint and knuckle joint.
- f) Write the name of the most common limit gage used for inspecting the hole diameter.
- g) What do you mean by 'allowance' in limits and fits?
- h) What is a key?
- i) Socket is used in which joint?
- j) Name the three parts of a pulley.
- k) Show the V-weld symbol.
- l) Sugarcane Juice maker is an example of which type of fit?
- m) Which type of coordinate system is used in AutoCAD?
- n) Show the symbol of ply wood section in Engineering Drawing.
- o) Name three types of rigid shaft couplings.

Group B

4. Answer the following questions (any six):

2x6

- i. Define taper with example.
- ii. What is the principle of development of surfaces?
- iii. Draw a sketch of interference fit.
- iv. Which method of development is employed in case of cones?
- v. In '50H7/c8' - what is represented by the number 50?
- vi. Draw (a) centre lines (b) visible lines (c) hidden lines.
- vii. What is an oblique cylinder?
- viii. Why the face of a pulley is crowned?
- ix. What are the basic requirements of a part drawing?
- x. What are the three classes of fits as per BIS system?

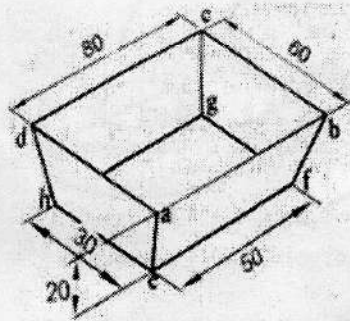
Group C

1x6

5. Answer the following question (any one):

a) A vertical square prism, base 40 mm side, is completely penetrated by a horizontal square prism, base 25 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the VP. while the faces of the two prisms are equally inclined to the VP. Draw the projections of the solids, showing lines of intersection. Taking lengths for the prisms as 50mm.

b) Draw the development of the surfaces of the given tray.



- c) Give a neat sketch of two cylinders being penetrated with each other at their axes at right angle.

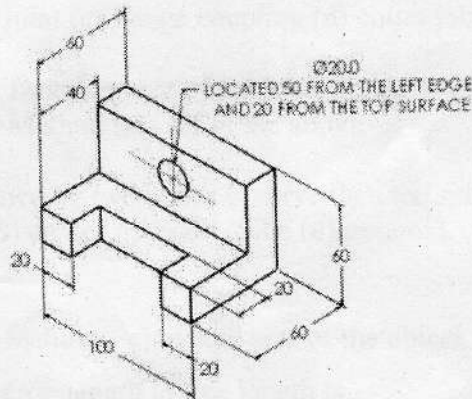
6

6. Answer the following question (any one):

1x6

- a) Draw the orthographic views (front and top) from the Isometric view.

6



- b) What is meant by hole basis and shaft basis system? Which one is preferred and why?

 $3+3$

- c) What is Thread terminology? Explain it with a suitable figure.

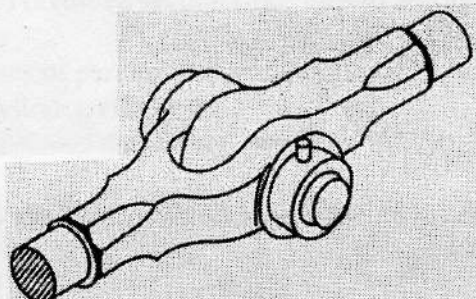
 $2+4$

7. Answer the following question (any one):

1x6

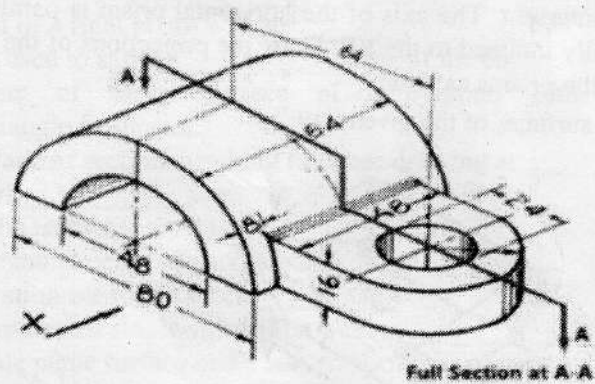
- a) Make neat freehand sketch orthographic view of each part of the following assembly of a knuckle joint (only elevation) and name them.

6



b) Draw the profile of involute teeth for a spur gear having 25 teeth and module of 10 mm, and assuming pressure angle of 20° . (Draw at least 3 teeth) 6

c) Draw the Full sectional elevation (i.e., section at A-A) viewed from X and right-hand side view as shown in figure. 6



March 2021

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and Answer any three questions from the rest.

1. Answer the following questions (any eleven):

11x1

State whether the following statements are True or False:

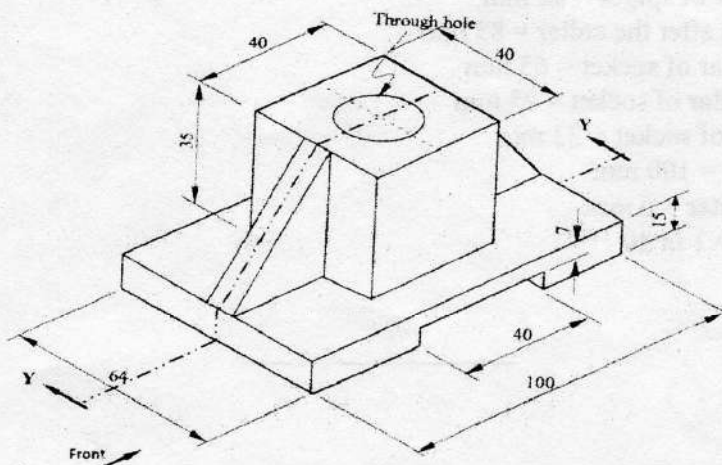
- i) A knuckle joint is used for rods which are in a straight line.
- ii) BSW thread means Bureau of Standard Whitworth thread.
- iii) The 'v' threads have more friction and are stronger than the square threads.
- iv) Grub screw is used to prevent the rotation of brasses in a bearing.
- v) Screw jack has square thread.
- vi) The big end of a connecting rod is the end fitted to the gudgeon pin.
- vii) Involute of a circle is used as the profile of a gear tooth.

Fill in the blanks:

- viii) _____ joint is used for rods in tension or compression.
- ix) _____ is the radial height of a tooth above the pitch circle.
- x) The number of units of PCD per teeth is called _____.
- xi) In case of saddle key _____ keyway is provided on the shaft.
- xii) A machine spindle can be driven at different speeds by fixing _____ on the spindle.
- xiii) Foundation bolts are used for _____ machines to their foundations.
- xiv) _____ bearing is used to supports for the vertical rotating shaft.
- xv) All lines and curves in the development represent only the _____ dimensions.

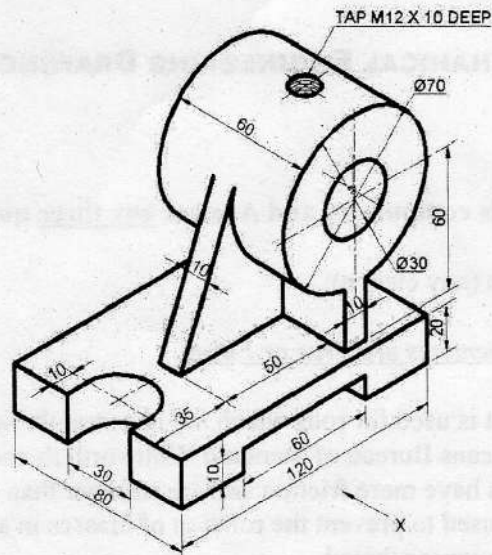
2. Draw a sectional front, top and left hand view of following diagram.

8

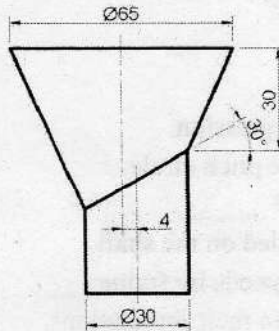


- 3. A vertical square prism, base 40 mm side, is completely penetrated by a horizontal square prism, base 25 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the V.P. While the faces of the two prisms are equally inclined to the V.P. draw the projections of solids, showing lines of intersection. Assume suitable lengths for the prisms. 8
- 4. Draw neatly and proportionately a spur gear of 20 teeth, P.C.D. = 300 mm. pressure angle = 20° , showing at least three nos. of teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle, face, flank, etc. 8

Draw full sectional elevation & plan of the following isometric view. The mark 'X' indicates front side. 8



6. Develop the lateral surfaces of these transition pieces. 8



7. Draw only the sectional elevation of a socket & spigot joint considering the following dimensions: 8

- Dimensions of both rods = 25 mm
- Diameter of collar of spigot = 50 mm
- Thickness of collar of spigot = 15 mm
- Outside diameter of spigot = 32 mm
- Length of spigot after the collar = 85 mm
- Diameter of collar of socket = 65 mm
- Thickness of collar of socket = 25 mm
- Inside diameter of socket = 32 mm
- Length of cotter = 100 mm
- Thickness of cotter = 6 mm
- Taper of cotter = 1 in 30

December 2019

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.

Answer any three questions from the rest.

1. A. Fill in the blanks (attempt all):

1x6

- i) A screw is specified by _____ diameter.
- ii) In gear circular pitch $\times$ No. of teeth = _____ $\times \pi$.
- iii) Pitch circle is _____ than base circle.
- iv) Bench vice has _____ thread.
- v) A small size screw external thread is often cut by means of a tool called _____.
- vi) Addendum circle is _____ than dedendum circle.

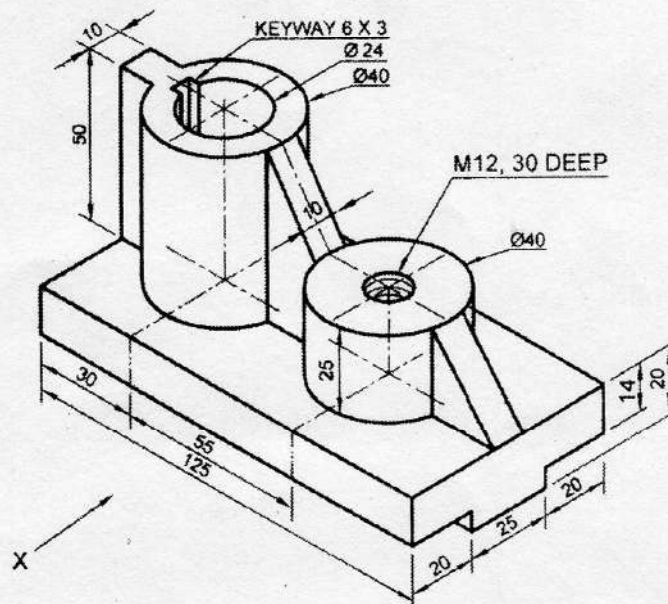
B. State whether following statements are True or False:

1x5

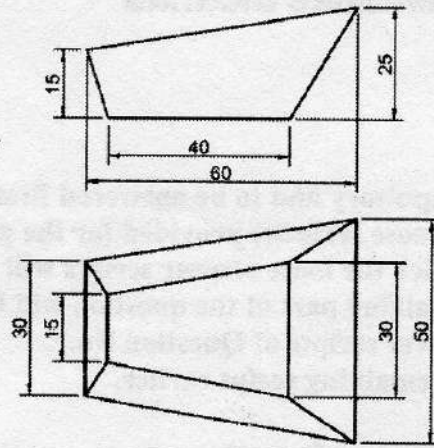
- i) Section lines are drawn approx. 2 mm gaps.
- ii) Module is the ratio of No. of teeth to PCD.
- iii) Pressure angle of involutes spur gear is usually taken as 20° .
- iv) The ratio of isometric length and true length is $\sqrt{2} : \sqrt{3}$
- v) Pressure angle is an angle between line of action and common tangent to the pitch circle.

2. A right cylinder of base $\phi 70$ mm and axis 100 mm long is completely penetrated by another cylinder of $\phi 40$ mm. the axis of the smaller cylinder is parallel to both the HP & VP and intersects at the midpoint of the bigger cylinder. Draw projections showing curves of intersection. 8

3. Draw only full sectional elevation of the following isometric view. The mark 'X' indicates front side. 8



4. A single stainless-steel sheet is to be cut into some pieces to fabricate a sheet metal tray. Develop the lateral surfaces of the following figure of the tray. 8



5. Draw neatly and proportionately a spur gear of 25 teeth, P.C.D. = 600 mm. pressure angle = 20° , showing at least three nos. of teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle, face, flank, etc. 8
6. Draw elevation, plan and side view of a hexagonal headed bolt & nut with a washer adopting the following dimensions. 8
- Diameter of bolt (d) = 20 mm
 - Length of shank = 100 mm
 - Thickness of head of the bolt = $0.8d$
 - Length of the threaded part of the bolt = 75 mm
 - thickness of the nut = d
 - Width across flats of hexagonal bolt and nut = $1.5d + 3$ mm
 - Width across corners of hexagonal bolt and nut = $2d$
 - Angle of chamfer = 30°
 - Radius of chamfer arc = $1.4d$
 - Diameter of washer = $2d + 3$ mm,
 - Thickness of washer = $0.12d$

December 2018

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.

Answer any three questions from the rest.

One mark is reserved for Draftsmanship.

1. Answer the following questions (any ten):

1×10

Fill in the blanks:

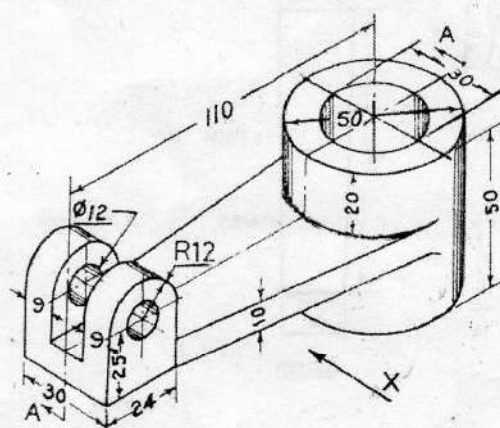
- i) A _____ coupling permits within certain limits, relative rotation and variation in the alignment of shafts.
- ii) The three parts of a pulley are hub, arms and _____.
- iii) _____ is the number of teeth per unit length of P.C.D.
- iv) The curve along which the two lateral surfaces meet each other is known as _____.
- v) Sphere and ellipsoid are the _____ curved surface solids.
- vi) When axis of any solid is inclined to its base, the solid is called _____ solid.
- vii) _____ method is applied to develop the lateral surfaces of transition pieces and oblique pyramid.

State whether the following statements are True or False:

- viii) Buttress thread is a combination of triangular and square threads.
- ix) A cotter is used to connect rigidly two rods subjected to shear.
- x) A knuckle joint is used for joining D-slide valve and eccentric rod of a steam engine.
- xi) In the fast and loose pulley, the fast pulley is free to rotate on the shaft.
- xii) In journal bearing the bearing pressure is parallel to the axis of the shaft.
- xiii) Bushes are prevented from rotation by using snug at the bottom.
- xiv) Engine cross head acts as a connecting link between the piston rod and the connecting rod.

2. Draw sectional front view and top view of the following Fig.1.

8

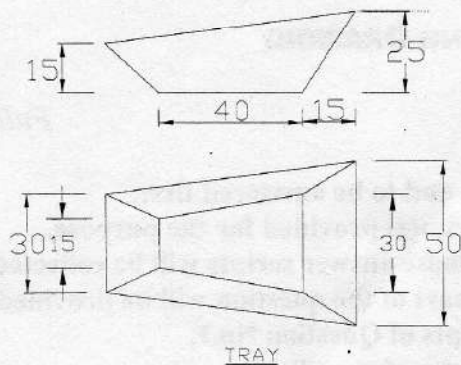


Front

Fig.1

3. Draw the lateral surface development of a tray shown in Fig.2.

8



4. A vertical cylinder of 70 mm diameter and 120 mm long resting on HP with one of its bases. It is completely penetrated by a square prism of 40 mm side of base and 120 mm long. The axes bisect each other at right angle. Draw the curves of intersection when all the base sides of prism are equally inclined to HP. 8
5. Draw the profile of an involute teeth for a spur gear having 30 teeth and module equal to 10 mm. Assuming a pressure angle of 20° . 8
6. Draw full sectional front view and side view of a C.I. pulley with the following dimensions. 8
- Bore dia = 22mm
 - Hub dia = 45 mm
 - Length of hub = 36 mm
 - Dia of pulley = 160mm
 - Width of pulley rim = 40mm
 - Thickness of pulley rim = 5mm
 - Number of arms = 4
 - Taper arm at boss = 20mm $\times$ 12mm and at rim = 12 mm $\times$ 6 mm
 - Crowning = 1.5mm
 - Key 6 mm $\times$ 4 mm
- Missing data, if any may be assumed.
7. Draw the full sectional front view and side view of a protected type rigid shaft coupling having diameter of shaft is 100 mm. Take standard proportions for other dimensions. 8
8. Fig-3 shows the details of spigot and socket joint. Assemble the parts and draw the sectional front view and side view. 8

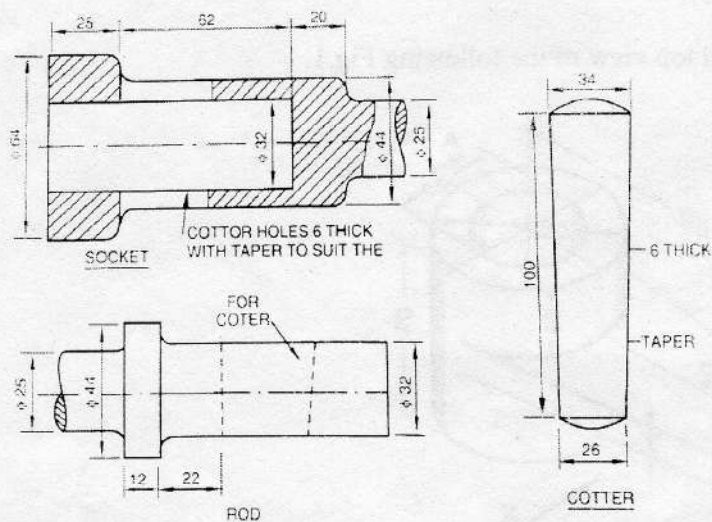


Fig-3